



Joowon Kim김주원

Full-Stack Engineer | Blockchain Engineer

Next.js · NestJS · Java · PostgreSQL · BSC/EVM

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SUMMARY

Hands-on full-stack and blockchain engineer with 6+ years of experience building production web applications, BSC/EVM wallets, stablecoin payment flows, smart contracts, token platforms, and AI-powered SaaS. Experienced across React/Next.js frontends, NestJS and Spring Boot backends, PostgreSQL/Redis data systems, asynchronous workers, wallet security, and on-chain transaction integration. Able to own architecture, implementation, deployment, and production operations while translating ambiguous business requirements into reliable software.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Java, Solidity, SQL
Frontend	React, Next.js, Tailwind CSS, responsive web applications
Backend	Node.js, NestJS, Spring Boot, REST APIs, event-driven processing
Blockchain	BSC/EVM, ethers.js, WalletConnect, Web3j, Hardhat, BEP-20, BscScan/Etherscan APIs
Data & Infrastructure	PostgreSQL, Redis, Prisma, BullMQ, Docker, AWS
Security & Quality	OTP, WebAuthn/Passkeys, encrypted key storage, schema validation, audit logging

EXPERIENCE

Blockchain Product LeadUNIDECAMay 2025 – Present

Seoul, South Korea · Hands-on full-stack and blockchain engineering lead

- Architect and implement **five production BSC wallet and token platforms** using React/Next.js, NestJS, PostgreSQL, Redis, ethers.js, WalletConnect, and BscScan integrations.
- Designed hybrid settlement systems that verify USDT deposits and withdrawals on-chain while processing purchases, staking, lockups, rewards, and transfers through auditable off-chain ledgers.
- Built wallet security and operational controls including encrypted key storage, OTP, Passkeys/WebAuthn, automated transaction verification, Telegram event alerts, treasury reconciliation, gas-reserve monitoring, and administrator audit logs.
- Developed smart-contract purchase flows with BEP-20 approvals, receipt and event verification, transaction synchronization, configurable pricing, treasury controls, pausing, and ownership management.
- Launched and operate a hybrid BSC platform serving **783 registered users and 398 purchasers**, processing **844K USDT in cumulative token purchase volume** within five months.

Blockchain Product Manager **TrillionsLab**

Aug 2021 – Oct 2024

Smart contract development · Java platform integration

- Designed and developed Solidity contracts for Emetal Exchange, a government-backed non-ferrous-metals marketplace supporting standard and reverse auctions, escrow settlement, winner payment, and unsuccessful-bid refunds.
- Connected smart contracts to the existing Java platform using MetaMask, Web3j, Infura, and Etherscan transaction tracking, with automatic retry and transaction-state handling.
- Coordinated independent validation that achieved **100% smart contract execution accuracy** and supported 200 concurrent users with a 0% server error rate.
- Contributed to the patented “Integrated Aluminum Ingot Trading Management System” and supported BlockApps’ Korea market entry through direct technical coordination with its U.S. team.

Java Backend Developer **Delta Dental of Michigan**

Feb 2020 – Jan 2021

- Developed Java/Spring Boot REST APIs and JDBC database integrations for an enterprise dental insurance intranet within a distributed Agile engineering team in Lansing, Michigan.

Software Engineer Intern **Apolis**

Aug 2019 – Jan 2020

- Built Java/Spring Boot REST APIs for a microservice-based hotel reservation management system, working with relational databases, backend architecture patterns, and unit testing.

SELECTED ENGINEERING PROJECTS

VIORA — Global Medical & Beauty Platform

Jun 2026 – Present

Turborepo · Next.js 15 · React 19 · NestJS 11 · Prisma · PostgreSQL · OAuth2 · Stripe · Cloudflare Workers

- Architected and launched a multi-tenant platform comprising four applications: customer web, hospital administration, super-admin console, and backend API, managed in a Turborepo/pnpm monorepo.
- Implemented JWT authentication and seven OAuth2 providers—Google, Kakao, Naver, Facebook, LINE, Apple, and WeChat—including JWKS-based token verification.
- Integrated Stripe payments and designed deposit and settlement automation; built booking/status workflows, product variants, rewards, coupons, real-time chat, web push, reviews, QR confirmation, audit logs, approval workflows, and CMS tools.
- Delivered a multilingual PWA on Cloudflare Workers using OpenNext and implemented GA4 event tracking for sign-up, login, checkout, and purchase.

RankFit — AI-Powered SEO / AEO / GEO Intelligence SaaS

May 2026 – Present

Next.js · NestJS · Prisma · PostgreSQL · Redis · BullMQ · Playwright · PageSpeed · OpenAI · Claude

- Designed and built a multi-page crawling and analysis platform validated across **19 web projects**, combining deterministic scoring, competitor benchmarking, structured diagnosis, and AI-search visibility testing.
- Architected asynchronous crawl and LLM workloads with BullMQ/Redis and implemented Zod schema validation, prompt-injection safeguards, data redaction, retries, and graceful fallbacks.
- Guided a live client implementation that improved the total score from **77 to 89**, including Technical SEO 87→100, On-page 82→100, AEO 24→74, and GEO 58→85 under the same scoring model.

Hybrid BSC Wallet & Token Platform

Oct 2025 – Present

React · NestJS · PostgreSQL · Redis · ethers.js · WalletConnect · BscScan · WebAuthn

- Built a production wallet integrating external MetaMask and Binance Wallet accounts through WalletConnect, automated USDT deposit verification, off-chain token accounting, and on-chain withdrawal settlement.
- Implemented staking and lockup engines, purchase and withdrawal queues, Passkey/OTP security, Telegram alerts, BscScan transaction links, multilingual UI, simulations, and detailed administrator workflows.
- Built treasury dashboards for fund flows, gas reserves, token balances, ledger-to-on-chain reconciliation, external balance adjustments, commissions, margins, and administrative audit history.

CourseNow — Cross-Platform Golf Community

Jul 2026 – Present

Next.js 15 · React 19 · TypeScript · Capacitor · Supabase · PostgreSQL · RLS · FCM · APNs · Playwright

- Built and launched a shared-codebase web, iOS, and Android application using Next.js and Capacitor, with Supabase/PostgreSQL, Naver OAuth, and Row-Level Security.
- Implemented round and score tracking, golf-course discovery, reports, reviews, rankings, a credit economy, cryptographically verifiable raffles, a point shop, and administration dashboards.
- Delivered native push through FCM/APNs plus PWA web push, optimized multi-resource fetching to avoid N+1 queries, and added WASM-based face blurring for uploaded images.
- Established Playwright E2E tests, Node.js integration tests, SQL migration management, and a GitHub Actions CI/CD pipeline.

EDUCATION

University of Minnesota, Twin Cities

Sep 2017 – May 2019

B.A., Computer Science (major) · Mathematics (minor) — GPA 3.0/4.0 · Transfer student

LANGUAGES

Korean (native) · English (fluent) · Extensive U.S. academic and professional experience